

Member ID: _____

Time: _____

Rank: _____



ADVANCED SPREADSHEET APPLICATIONS (235)

NATIONAL 2022

Production:

Job 1: Worksheets with answers and formulas	_____ (295 points)
Job 2: Formatted worksheet	_____ (115 points)
Job 3: Clustered column chart	_____ (35 points)
TOTAL POINTS	_____ (445 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. Put your Member ID and printout number in the right section of the footer on each printout unless instructed otherwise. Your name or initials should *not* appear on any work you submit.
5. If you finish before the end of the testing time, notify the proctor. Time may be a factor in determining the winner in the event of a tie.
6. When turning in your contest, the jobs should be arranged in printout order.

SCENARIO:

You work for Handy Handyman construction company. Your company has done a complete bathroom remodel and switching of a tub/shower combo to just a shower for Brandy McClain. Today you have to complete the bathroom remodel invoice summary for Brandy McClain with the total costs, which includes how much she paid with her down payment. You need to create and input the information into the invoice spreadsheet, compute the costs, and send to her by email. Then you need to create another worksheet that shows the bathroom remodel comparisons from actual cost to invoice costs, it must show where it cost you more money than you invoiced and if you lost any money on the job. Based on the comparison create a chart.

Job 1 (Worksheets with answers and formulas)	Points Possible	Points Earned
Member ID and Job # in right section of footer	10	
Calibri Size 11 font used for data	10	
Landscape orientation with narrow margins	10	
Worksheet printed landscape on one page centered vertically and horizontally	10	
Type in A1 Handy Handyman	5	
Type in A2 Remodeling Invoice	5	
Type in A3 Items	5	
In B43:C43 a function that calculates the total material costs	10	
Copy A1:B3 to the “Bathroom Remodel Summary” worksheet	5	
Enter a function in A5 that references cell A5 from the original worksheet	5	
Enter a function in A6 that references cell A15 from the original worksheet	5	
Enter a function in A7 that references cell A18 from the original worksheet	5	
Enter a function in A8 that references cell A21 from the original worksheet	5	
Enter a function in A9 that references cell A25 from the original worksheet	5	
Enter a function in A10 that references cell A29 from the original worksheet	5	
Enter a function in A11 that references cell A33 from the original worksheet	5	
Enter a function in A12 that references cell A36 from the original worksheet	5	
Enter a function in A13 that references cell A39 from the original worksheet	5	
Enter a function in B5 that calculates the sum of the Bath_Shower range	5	
Enter a function in B6 that calculates the sum of the Cabinets range	5	
Enter a function in B7 that calculates the sum of the Countertops range	5	
Enter a function in B8 that calculates the sum of the Faucets range	5	
Enter a function in B9 that calculates the sum of the Flooring range	5	
Enter a function in B10 that calculates the sum of the Lighting range	5	
Enter a function in B11 that calculates the sum of the Sinks range	5	
Enter a function in B12 that calculates the sum of the Toilets range	5	
Enter a function in B13 that calculates the sum of the Walls range	5	
Add title and enter a formula in C15 that totals the labor and materials costs	10	
Add title and enter an absolute formula in C16 that calculates the incidental costs	10	
Add title and enter a formula in C17 for the Job Total	10	
Add title and enter a formula in C19 that calculates the 20% down payment	10	
Add title and enter a formula in C20 that calculates the total due	10	
Copy the worksheet	5	

Delete rows 15-20	5	
Update worksheet with titles and Actual Cost	10	
Enter in formula in D5:D13 that calculates the difference between the total and actual costs	10	
Enter function to calculate the totals in B15:D15	5	
In B15 place a function that totals the total cost column and fill this function over to C15 and D15	10	
In D16 use a reference function to pull the total incidentals information	5	
In D17, insert a formula that calculates the Job Loss/Job Covered by the amount budgeted for incidentals	5	
In cell D18 place a function that finds the highest difference of loss	5	
In cell E3, add the title, "Over/Under". Add a nested If statement in E5 that calculates if the items in over or under budget. Copy the If statement from E5 to E13	20	
Print a copy of the worksheet with gridlines and adjust column widths to make all data visible and print with the formulas showing	5	
Subtotal	295	
Job 2 (Formatted worksheet)	Points Possible	Points Earned
Member ID and Job # in right section of footer	10	
Original Sheet: Merge and center the title A1 through C1.	5	
Merge and center the subtitle A2 through C2.	5	
Format the company name to Title cell style.	5	
Format the subtitle to Heading 4 cell style.	5	
Center and Bold A3:C3	5	
Format columns B and C to comma with 2 decimal places.	5	
Increase column width of all columns using auto fit.	5	
Bathroom Remodel: Merge and center the title A1 through C1.	5	
Merge and center the subtitle A2 through C2.	5	
Format the company name to Title cell style.	5	
Format the subtitle to Heading 4 cell style.	5	
Center and Bold A3:B3	5	
Format B5:C20 to comma with 2 decimal places.	5	
Comparison: Merge and center the title A1 through E1	5	
Merge and center the title A1 through E1	5	
Merge and center the subtitle A2 through E2	5	
Format the company name to Title cell style	5	
Format the subtitle to Heading 4 cell style	5	
Center and Bold A3:E3	5	

Format B5:D18 to comma with 2 decimal places	5	
Print each spreadsheet with all data visible, centered, fit to one page	5	
Subtotal	115	
Job 3 (Clustered Column Chart)	Points Possible	Points Earned
Create a clustered column chart	10	
Select the chart style 2	5	
Title of the chart is "Cost Comparison"	5	
Title is Impact font size 28.	5	
Make sure the legend is showing at the top with data labels at the outside end	5	
Print the chart on a separate sheet	5	
Subtotal	35	
TOTAL POINTS	445	

JOB 1: Create Spreadsheet with Formulas and functions

1. Set the global options to include the following modifications: Calibri, 11pt, Landscape, Centered horizontally and vertically, Fit to One Page, and Narrow Margins.
2. Using the Handyman worksheet provided, type in A1 enter “Handy Handyman”, A2 enter “Remodeling Invoice”, and in A3 with “Items”.
3. In B43, place a function that calculates the total material costs. Fill the function to C43.
4. Define the following ranges of information:
 - B6:C13 as Bath_Shower
 - B16:C16 as Cabinets
 - B19:C19 as Countertops
 - B22:C23 as Faucets
 - B26:C27 as Flooring
 - B30:C31 as Lighting
 - B34:C34 as Sinks
 - B37:C37 as Toilets
 - B40:C40 as Walls
5. Create a new worksheet and name it “**Bathroom Remodel Summary**”.
6. Copy A1:A3 over from the “Bathroom Remodel Breakdown” into A1:B3 of “Bathroom Remodel Summary”.
7. Enter “Total Cost” in B3.
8. Enter a function in A5 that references A5 from the original worksheet.
9. Enter a function in A6 that references A15 from the original worksheet
10. Enter a function in A7 that references A18 from the original worksheet
11. Enter a function in A8 that references A21 from the original worksheet
12. Enter a function in A9 that references A25 from the original worksheet
13. Enter a function in A10 that references A29 from the original worksheet
14. Enter a function in A11 that references A33 from the original worksheet
15. Enter a function in A12 that references A36 from the original worksheet
16. Enter a function in A13 that references A39 from the original worksheet
17. Enter a function in B5 that calculates the sum of the Bath_Shower range.
18. Enter a function in B6 that calculates the sum of the Cabinets range.
19. Enter a function in B7 that calculates the sum of the Countertops range.
20. Enter a function in B8 that calculates the sum of the Faucets range.
21. Enter a function in B9 that calculates the sum of the Flooring range.
22. Enter a function in B10 that calculates the sum of the Lighting range.
23. Enter a function in B11 that calculates the sum of the Sinks range.
24. Enter a function in B12 that calculates the sum of the Toilets range.
25. Enter a function in B13 that calculates the sum of the Walls range.
26. In A15, enter the text “Labor and Materials”. Enter a formula in C15 that references from the totals the labor and materials costs together.
27. In A16, enter the text “Incidental”. Enter an absolute formula in C16 that calculates the incidental costs at 10% against the labor and materials costs. The 10% is in cell B16 and should be absolute.
28. In A17, enter the text “Job Total”. Enter a formula in C17 that adds the incidental costs to the labor and materials total costs.

29. In A19, enter the text “Down Payment”. Enter a formula in C19 that calculates the 20% down payment on the job total cost. The 20% is in B19 and should be absolute.
30. In A20, enter the text “Total Due”. Enter a formula in C20 that calculates the total due.
31. Copy the worksheet and rename it **Comparison**.
32. Delete rows 15 to 20.
33. Update the worksheet with the following information including the Actual Cost column.

Handy Handyman			
Remodeling Invoice			
Items	Total Cost	Actual Cost	Difference
Bath/Shower		4585	
Cabinets		630	
Countertops		450	
Faucets		1500	
Flooring		1380	
Lighting		480	
Sinks		885	
Toilets		988	
Walls		1100	
Difference			
Incidentals			
Job Loss/Job Covered			
Highest Difference of Loss			

34. Enter in formula in D5:D13 that calculates the difference between the total and actual costs.
35. In B15:D15 place a function that totals the total cost column.
36. In D16 use a reference function to pull the total incidentals information from Bathroom Remodel Summary worksheet.
37. In D17, insert a formula that calculates the Job Loss/Job Covered by the amount budgeted for incidentals.
38. In cell D18 place a function that finds the highest difference of loss, remember there are negative numbers that indicate a loss.
39. In cell E3, add the title, “Over/Under”. Add a nested If statement in E5 that calculates if the items in over or under budget. Copy the If statement from E5 to E13.
40. Print a copy of the worksheet with gridlines and adjust column widths to make all data visible, **formulas showing**.

JOB 2: Format the Spreadsheet you created in Job 1 using the following instructions:

1. Merge and center the title A1 through C1.
2. Merge and center the subtitle A2 through C2.
3. Format the company name to Title cell style.
4. Format the subtitle to Heading 4 cell style.
5. Center and Bold A3:C3
6. Format columns B and C to comma with 2 decimal places.
7. Increase column width of all columns using auto fit.
8. Print the worksheet with all data showing, fit to one page.

9. Move to worksheet: **Bathroom Remodel Breakdown.**
10. Merge and center the title A1 through C1.
11. Merge and center the subtitle A2 through C2.
12. Format the company name to Title cell style.
13. Format the subtitle to Heading 4 cell style.
14. Center and Bold A3:B3
15. Format B5:C20 to comma with 2 decimal places.
16. Print the worksheet with all data showing, fit to one page.

17. Move to worksheet: **Comparison.**
18. Merge and center the title A1 through E1.
19. Merge and center the subtitle A2 through E2.
20. Format the company name to Title cell style.
21. Format the subtitle to Heading 4 cell style.
22. Center and Bold A3:E3
23. Format B5:D18 to comma with 2 decimal places.
24. Print the worksheet with all data showing, fit to one page.

JOB 3: Create a Clustered Column Chart

Create a clustered column chart from the Comparison you have created in Job 2. Move the chart to its own sheet. Format the chart to look like the following.

- Select the chart style 2
- Make sure the title of the chart is “Cost Comparison.”
- Title is Impact font size 28.
- Make sure the legend is showing at the top with data labels at the outside end.
- Print the chart.

You should have the following printouts:

Job 1—Three (3) worksheets printed with formulas

Job 2—Three (3) worksheets formatted

Job 3—Clustered Column Chart